SMARTBRIDGE
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SKILL WALLET
A SmartBridge Product

Coding & Solution

Date	1 July 2026
Team ID	
Project Name	Human Development Index (HDI) Prediction – A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Coding & Solution

The Human Development Index (HDI) Prediction System was developed using Python, Scikit-learn, Flask, HTML, and CSS. It predicts the HDI based on key indicators such as Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and GNI per Capita. The trained Linear Regression model is integrated with a Flask web application, providing users with an interactive interface to obtain HDI predictions and the corresponding development category.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Santhoshi003/Human-Development-Index-Predictor
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	Flask, Scikit-learn, NumPy, Pandas
Key Features Implemented	• Data preprocessing and visualization. • Linear Regression model training. • HDI prediction based on user input. • Classification into Low, Medium, High, and Very High Human Development. • Interactive Flask web application with user-friendly interface.
Pending / Incomplete Features	• Country-wise comparison. • User authentication. • Historical trend visualization. • Deployment on a cloud platform.
Setup / Run Instructions	1. Install required Python libraries using pip install. 2. Place the trained HDI.pkl model in the project folder. 3. Run python app.py. 4. Open http://127.0.0.1:5000 in a web browser. 5. Enter the required values and click **Predict HDI**.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Partial
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

- The project follows a modular structure with separate folders for the Flask backend, HTML templates, CSS files, and the trained machine learning model.
- A Linear Regression algorithm is used to predict the Human Development Index accurately.
- The web application provides an intuitive interface for users to enter input values and instantly view the predicted HDI and its development category.
- The project can be further enhanced by adding data visualization dashboards, cloud deployment, and support for real-time HDI datasets.